

Post-Quantum Mathematics

Part 1 - Standards Tracks & Candidates: NIST Standards Track

Compiled by the Spaced Research Ledger

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NIST runs the process that decides which post-quantum algorithms the world will actually use. Its choices are not binding on anyone outside the US federal government, but in practice they set the direction for browsers, cloud providers, hardware vendors and most other standards bodies.

Three algorithms are finished and published. A fourth is in final review. A fifth was selected but not yet written up. A further nine are still competing in a separate process for additional signature schemes. This part covers all four of those stages, because the difference between them is the difference between something you can deploy and something you should only watch.

One piece of vocabulary carries through. A KEM, or key encapsulation mechanism, is the modern replacement for classical key exchange, and it is how two parties agree on a secret over an open channel. Signature schemes prove who sent something. The two have different urgencies, for reasons Part 2 covers.

Section 1 covers the three ratified standards and the corrections being made to them. Section 2 covers the signature standard still in review. Section 3 covers the additional-signatures competition. Section 4 covers HQC. Section 5 covers the IETF work that turns these standards into usable protocols.

1. The Ratified Standards

Three standards were published together in August 2024 and are the settled foundation of the whole migration. What is interesting about them now is not their content but their errata, because both lattice-based standards are being corrected while already deployed.

Recent Developments

NIST's FIPS 204 publication page carries a planning note dated July 2026, pointing to an errata spreadsheet listing several minor issues to be corrected in a future revision [1]. The spreadsheet's specific contents were not retrieved for this report.

Checked as of the date of this report, the FIPS 205 page carries no planning note and no errata link [2]. Its document history lists only the original draft and final entries. Of the three ratified standards, the hash-based one is the only one with nothing queued for correction.

Two pieces of validation infrastructure moved in August 2026, which matters to anyone certifying an implementation rather than reading the standard.

The FIPS 140-3 Implementation Guidance was updated on 19 August 2026 [3]. It adds clarifications on hybrid modules, new scenarios for hybrid key encapsulation referencing SP 800-227, and updated self-test

requirements for ML-KEM. Hybrid deployment is the norm in the migration, so guidance on how hybrid modules are validated is the practical gate.

Draft validation-testing specifications for ML-DSA and SLH-DSA were published dated 14 August 2026 [4]. They cover known-answer tests, signature verification test types and coverage details. Testing specifications are what turn a published standard into something a laboratory can certify against.

Current State

NIST published the first three post-quantum standards on 13 August 2024: FIPS 203 for ML-KEM, FIPS 204 for ML-DSA, and FIPS 205 for SLH-DSA [5].

FIPS 203 has an open issue. NIST's own publication page carries a planning note dated November 2025 [6]. It states that a problem has been identified which will be corrected in a future revision, with details in a linked errata spreadsheet. The spreadsheet itself was not retrieved for this report.

A mailing-list thread on the NIST PQC forum discusses two specific planned changes to FIPS 203 [7]. The first allows the key-derivation methods in SP 800-56C to apply to ML-KEM-generated keys in place of shared secrets. The second specifies a lower-level derandomized interface, which enables validation testing against known answers and allows keys to be stored as compact seeds rather than in expanded form. Seed storage matters for constrained devices, where ML-KEM's key sizes are a real cost.

FIPS 204's page also carries an earlier planning note from January 2025, announcing a general FAQ resource covering the post-quantum standards rather than anything specific to that document [2].

2. FN-DSA and FIPS 206

One algorithm selected in 2022 still has no published standard. The reason is instructive: the difficulty is not in the mathematics but in implementing it safely.

Current State

FN-DSA, previously called Falcon, was left out of the initial batch of three because its reference signing procedure depends on floating-point Gaussian sampling [8]. That is difficult to implement safely and in constant time, and it is the main reason its standardization has taken longer than the others. Constant-time implementation matters because an operation whose duration depends on the secret key leaks that key to anyone who can measure timing.

NIST submitted the draft FIPS 206 standard for approval in August 2025 [8]. As of mid-2026 it remains in final review, with the final standard expected in late 2026 or early 2027.

The algorithm is worth the wait for specific uses. FN-DSA produces the smallest signatures and public keys among NIST's post-quantum signature schemes, which makes it attractive wherever bandwidth is constrained [9]. NIST project staff have given conference presentations tracking the standard's status. This report could not extract readable text from the presentation itself, so that citation supports the presentation's existence and topic rather than any figure inside it.

As of this research, FN-DSA is something to plan and test for, not to deploy in production [8].

3. The Additional Signatures On-Ramp

NIST opened a second signature competition specifically to reduce its dependence on one kind of mathematics. Nine candidates remain, and the interesting fact about them is how few are lattice-based.

Recent Developments

NIST announced in May 2026 that nine candidates advanced from round 2 to round 3 [10]. One is lattice-based, HAWK. One is isogeny-based, SQIsign. Three are built on a technique called MPC-in-the-head: FAEST, MQOM and SDiTH. Four are multivariate: UOV, MAYO, QR-UOV and SNOVA.

The distribution is the point. Only one of the nine is lattice-based, against four multivariate schemes. The competition exists to find signatures whose security rests on different assumptions from ML-DSA's, so a future break in lattice mathematics would not take everything down at once.

SQIsign is the only isogeny-based scheme remaining, and has the smallest combined public-key-and-signature size of any round-3 candidate, at 148-byte signatures at the lowest security category [10].

Current State

Round-3 teams had to submit finalized specifications and implementation adjustments addressing known cryptanalytic issues by 14 August 2026, and NIST published its round-2 status report as NIST IR 8610 [10].

The evaluation period is expected to run roughly two years [10]. It concludes around the 7th NIST PQC Standardization Conference, tentatively scheduled for late spring or early summer 2027 in Gaithersburg, Maryland.

Part 5 covers what has happened to several of these candidates since. One of the nine has already been withdrawn.

4. HQC

NIST selected a fifth algorithm in 2025 for the same reason the signature competition exists: to avoid depending on one kind of hard problem. Its standard has not appeared.

Current State

NIST selected HQC as a fifth algorithm for standardization in March 2025 [11]. It is a code-based key encapsulation mechanism, chosen specifically to give the portfolio a backup built on different underlying mathematics from ML-KEM's structured lattices, should ML-KEM ever be broken.

NIST mathematician Dustin Moody stated the reasoning directly [11]. In his words, NIST wanted a backup standard based on a different mathematical approach from ML-KEM.

As of that announcement no FIPS number had been assigned [11]. A draft standard was expected roughly a year after selection, around March 2026, with a final standard anticipated in 2027.

Unresolved Conflicts

A secondary source states that HQC's final standard is "anticipated as FIPS 207", naming a specific number [12]. This report could not verify that number against any NIST primary source, and NIST's own selection announcement is silent on it [11]. The disagreement is between a specific claim and an absence rather than between two competing figures, which is why it is recorded rather than resolved. A NIST publication or draft carrying the number would settle it.

5. IETF Standardization Activity

A NIST standard describes an algorithm. It does not say how to use that algorithm inside a real protocol. That second job belongs to the IETF, and this section covers the drafts that do it.

Recent Developments

The IETF TLS working group's draft for pure ML-KEM in TLS 1.3 has moved past working-group revision into IESG Last Call, reached around 30 July 2026 [13]. Final comments were due 13 August 2026, with an intended status of Informational RFC. Last Call is the final review stage before publication, so this work is close to done rather than in progress.

The draft defines the three ML-KEM parameter sets as named groups for TLS 1.3 and registers the corresponding values in the public registry [14]. Registration is the step that lets independent implementations interoperate.

Current State

The LAMPS working group's composite-signatures work reached working-group last call in March 2026, targeting publication as an RFC in the final quarter of 2026 [15]. A composite certificate carries a single subject public key formed by joining an ML-DSA-65 public key to a classical ECDSA P-256 public key. The point of a composite is that an attacker must break both to forge a signature.

The HPKE working group has an active draft defining post-quantum and hybrid key encapsulation for hybrid public key encryption [16]. It covers the three ML-KEM parameter sets and hybrid combinations pairing ML-KEM-768 with classical X25519 or P-256.

About This Report

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